

ORIGAMI AS A SPATIAL AND GEOMETRIC REASONING BENCHMARK

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ABSTRACT

Multimodal language models are strong at tasks that ask what is in an image and weaker, in ways their image-task scores do not predict, at tasks that ask about geometry. We use origami to measure the difference. We release **CP2Seq**, a benchmark and dataset of 600 origami samples in which a model is given a crease pattern and the final folded state and must recover the sequence of folds between them. The task is trivial to generate and hard to solve: folding forward is one pass of an engine that records what it did, while deciding whether a crease pattern is reachable by simple folds is NP hard, and determining the layer ordering of a flat folding is NP hard even given a valid mountain-valley assignment. Ground truth is constructed rather than annotated, and every sample rebuilds byte-identically from its seed, so the corpus ships as a generator rather than as data. The environment the model works against is also the verifier: it executes one candidate fold and either returns the resulting state or refuses with a named reason, such as the sheet would tear. It performs no search, so proposing, pruning and backtracking stay with the model. We provide progressive tiers of tool assistance, and because one crease pattern admits many valid folded states we score answers by replaying them, up to the plane isometry group. At low reasoning effort no model we evaluated solved any sample beyond the easy stratum, and on that stratum a deterministic breadth-first search over the same action set solves 54.7% against the best model’s 20.4%. The dominant failure is not an exhausted budget but revisiting states already seen, which ends 48% of that model’s attempts.

1 POSITIONING

This paper is submitted as a **benchmark and dataset** contribution. What it offers the community is an evaluation substrate for **spatial and geometric reasoning in large language models**: a corpus, an environment that doubles as an exact verifier, and a scoring protocol, rather than a model, a training method or a state-of-the-art number. Every arm reported here runs an off-the-shelf model through the harness; nothing is trained. The claim under review is that the benchmark measures what it says it measures and that its judge can be trusted, not that any particular model is good at the task.

The reasoning being measured is geometric as much as it is spatial, and the distinction is load-bearing rather than cosmetic. A solver has to reason about lines, reflections, incidence and adjacency on the original sheet, which is geometry in the ordinary sense, and simultaneously about a layer ordering that the geometry does not determine, which is combinatorial. Calling the task spatial reasoning alone understates the second half. Positioning the work as a benchmark for spatial *and geometric* reasoning states the scope accurately and places it beside the multimodal-reasoning benchmarks of Section 3.1 rather than beside the computational-origami literature of Section 3.3, which is the theory the benchmark rests on rather than the field it contributes to.

2 INTRODUCTION

Multimodal language models have become genuinely good at visual work. They caption and answer questions about natural images, read documents, charts and tables, transcribe text in photographs, follow video, and drive graphical interfaces from screenshots alone. Over the last two years the

direction of travel on these tasks has been steeply upward, and on several of them the reported numbers sit near careful human annotators.

It does not follow that these models reason about space. Most of the competence arrives through one architecture: a vision encoder trained to align images with text, and an adapter that projects its output into the language model’s token stream. What that encoder is rewarded for preserving is semantic. It answers *what is in this picture*, and it answers it well, because that is what the training objective asks of it. Metric and relational structure is not rewarded and need not survive the projection: exact angles, exact incidences, which of two nearly identical shapes lies in front, how many times one line crosses another. Competence at recognising a scene and competence at reasoning about its geometry are separable, and an adapter trained for the first gives no guarantee of the second. When the task moves to geometric and spatial reasoning, the ordering of models by their image-task scores need not survive.

Origami is an unusually clean place to look for the missing capability. It is an art form whose entire content is geometric: a folder works from lines, reflections and incidences on a single uncut sheet, and every decision is constrained by every decision before it. It also demands spatial reasoning of a specific kind, because the sheet stops being flat the moment folding begins. The folder has to track where each piece of paper now sits, which pieces lie above which, and which of them are still joined to each other through the original sheet. A crease pattern, the flat record left behind when a finished model is unfolded, contains all of that history and none of its order. Reading a sequence back out of it is the skill that separates a practitioner from someone who can recognise a crane.

We use origami as an instrument. **The model is given two things: the crease pattern, and the final folded state. It is asked for the sequence of folds that turns the flat sheet into that state.** Both endpoints are supplied and the path between them is not, which is the whole of the task. The pattern says which lines were creased at some point during the folding but says nothing about the order they were creased in, and the final state says where the paper ended up but not how it got there.

This is trivial to generate and hard to solve, which is what makes it usable for evaluation. Folding forward is a single pass of an engine that records what it did. Deciding whether a crease pattern is reachable by simple folds at all is NP hard (Arkin et al., 2004; Akitaya et al., 2017), so recovering the sequence that produced one is at least as hard. The same engine that generates a sample therefore rules on any proposed step of a solution exactly, using an operation it already implements, with no learned component in the loop, no tolerance to tune and no oracle to trust.

That asymmetry is what a benchmark wants, and it is what spatial and physical reasoning has generally lacked. Benchmarks built on static images or multiple choice never execute the model’s proposal, so an answer that is right for the wrong reason is indistinguishable from one that is right. Benchmarks built on learned surrogate simulators do execute the proposal, but against an approximation whose wrong verdicts look exactly like its right ones, which leaves the benchmark’s own error rate unknown and unmeasurable from inside. Human judgement is exact in a sense and does not scale. The consequence is worth stating plainly, because it motivates everything below: **a benchmark can only be as trustworthy as the judge inside it.** Here the judge is the generator run forwards, so difficulty is controllable rather than sampled, ground truth is recorded rather than annotated, and negative examples are unlimited, because an illegal fold is produced by asking for one.

Two recent benchmarks make origami a target for multimodal evaluation, and the difference between them and this work is the difference between asking a model a question and executing its answer. **GamiBench** (Spencer et al., 2025) pairs crease patterns with folded shapes rendered from six viewpoints and scores multiple-choice visual question answering, including a set of deliberately impossible patterns a model ought to reject. **OrigamiSpace** (Xu et al., 2025) instruments each instance with a pattern diagram, the folding process and a final image, and defines four tasks over them, from pattern prediction to crease-pattern code generation. In both, a proposal is graded by comparison against a stored target: the model selects among rendered alternatives, or emits an artifact that is matched to a reference. Neither steps a paper-exact engine and asks whether *this* move, on *this* stack, is something paper could do. Both therefore measure recognition of folded geometry, which is a real capability, rather than the ability to search for a sequence that produces it. Section 3.4 makes the comparison in full.

The distinction matters because the search is where the difficulty lives. Determining the layer ordering of a flat folding is NP hard on its own, even when a valid mountain-valley assignment is supplied (Bern & Hayes, 1996). No amount of pattern recognition substitutes for search on a problem of that shape, and a benchmark that never executes a proposed step cannot tell a model that searched from a model that recognised, nor report how much search either one needed.

This is also why the benchmark reports a deterministic search baseline alongside the models rather than only against them. The stack deepens with every fold, the enumerated action set grows with it, and the depth at which exhaustive search stops being affordable is a property of the corpus that can be measured rather than asserted. [TO RUN: the deterministic breadth-first baseline of Section 9, reported per stratum, giving the depth at which it stops solving and the states it expands. Until that number exists, no claim is made here about where blind search fails.] What the benchmark is designed to reward is the use of geometry: looking at the crease pattern and the final state and inferring which fold could plausibly have been last, then working backwards. **Geometry is supplied to the model as input and is not the thing being scored. Being able to act on it is.** A model that cannot read a reflection off a pattern has no way to choose a direction other than enumeration, and the baseline is what says how far enumeration alone gets.

The task is also hard in ways a maze or a grid-world is not. It gets harder as it proceeds: every fold thickens the stack and constrains what the next fold may do, so the difficulty of step k depends on steps 1 through $k - 1$. And a solver has to maintain two kinds of state that constrain each other. The crease graph is fixed once the pattern is given, but vertex coordinates are reflected at every step, the overlap structure changes at every step, and the layer ordering changes at every step and is *chosen* rather than computed. Determining that ordering is NP hard even when a valid mountain-valley assignment is supplied (Bern & Hayes, 1996). Mazes have geometry alone; here an independent combinatorial layer is tracked alongside it. The action space is not invented for the occasion either: simple folds have been defined and their complexity studied for two decades (Arkin et al., 2004), so the operations an agent is given have names and claims about them can be checked against a literature.

We contribute the following.

1. **A dataset, and the generator that built it.** CP2Seq holds 600 origami samples, each pairing a crease pattern with the fold sequence that produced it, the folded state after every step, and per-sample difficulty metrics. Ground truth is constructed rather than annotated, because the sequence is recorded as the fold happens. Samples are generated under two named action models from the simple folding literature and stratified by fold depth and by coupling. The corpus ships as a generator and a manifest rather than as data: every sample rebuilds byte-identically from its seed, so the dataset can be regenerated, extended, or re-stratified at a different difficulty by anyone who wants a different corpus than ours.
2. **An evaluation framework and environment, with progressive tiers of assistance.** The model is given the crease pattern and the final state and must recover the sequence, so it works from partial information about the structure throughout. The environment it works against is also the verifier: stepping it is the same act as asking for a verdict. It executes one candidate fold and either returns the resulting state or refuses with a named reason, such as the sheet would tear or the fold creases nothing. It performs no search and makes no choices, so proposing, pruning and backtracking stay with the model. Around it sits a graded tool belt, from a full belt with rendered views down through withheld vision and raw pass/fail to no tools at all, so the contribution of each assistance level is measured rather than assumed. The framework also supplies the scoring protocol: answers are judged by replaying them through the engine, at two declared levels of equality, because one crease pattern admits many valid folded states and the recorded sequence is not guaranteed to be the shortest.
3. **An evaluation of frontier multimodal models** against this dataset through this framework, run across the progressive tiers of tool assistance and with a memorization control on anonymized geometry. [TO RUN]

4. **A measured scope boundary.** 89.3% of the 366 real crease patterns in Flat-Folder’s examples/instagram/ corpus (Flat-Folder) provably lie outside all-layers simple folding. The benchmark’s action space is bounded by evidence rather than by assertion, and the boundary is reported rather than buried.

We find that the task is beyond current models at the setting we could afford to run. No model solved a single sample past the easy stratum, and on the easy stratum a deterministic breadth-first search over exactly the action set the models are given solves 54.7% against 20.4% for the best-covered model. Blind search beats every model tested, on the only stratum where anything succeeds at all. The dominant failure is not exhausting the search budget but revisiting states already seen: 48% of that model’s attempts terminate in detected state cycling. Every model attempt ran at low reasoning effort, so the supported claim is about models at that setting rather than about the frontier in general; Section 11.6 states this and Section 12 carries it as a limitation.

3 RELATED WORK

3.1 ORIGAMI BENCHMARKS FOR MULTIMODAL MODELS

OrigamiSpace (Xu et al., 2025) contributes 350 instances, each carrying a crease pattern diagram, a compiled flat pattern, the complete folding process and a final folded image, and defines four tasks over them: pattern prediction, multi-step spatial reasoning, spatial relationship prediction, and end-to-end crease-pattern code generation. It also provides an interactive environment and describes reinforcement learning as a possibility explored rather than a result demonstrated. Its design lesson is that one richly instrumented sample can support several tasks, which is how 350 instances sustain a benchmark paper.

GamiBench (Spencer et al., 2025) contributes 186 regular and 186 *impossible* crease patterns paired with folded shapes from six viewpoints, across three visual question-answering tasks, and introduces two diagnostic metrics: viewpoint consistency and impossible fold selection rate. Its impossible patterns are the closest the current literature comes to forcing a model to judge feasibility rather than recognise a shape, and its headline finding is that leading models struggle even at single-step spatial understanding. It is the benchmark this work is most directly in conversation with, and the one whose negative examples we can produce without limit, because generating an illegal fold in our environment is the same operation as generating a legal one.

OrigamiBench (Agarwal et al., 2026), **FoldingAgent** (Moriya et al., 2026) and **COrigami** (Zahavy et al., 2026) complete the picture; their metrics appear in the comparison of Section 3.4.

What all of these share is the gap. In each, the model’s proposal is answered by a question rather than by an execution: it selects among rendered alternatives, or produces an artifact that is compared to a target. None of them steps a paper-exact engine and asks whether *this* move, on *this* stack, is something paper could do.

3.2 METHODS THAT PAIR A PROPOSER WITH A SIMULATOR

Learn2Fold (Huang et al., 2026) formulates origami as conditional program induction over a crease-pattern graph. A language model generates candidate folding programs from text, and a learned graph-structured world model acts as a differentiable surrogate simulator that predicts physical feasibility and failure modes before execution, inside a lookahead planning loop. Its stated key insight is to decouple semantic proposal from physical verification.

We adopt that decomposition and differ on one thing: the verifier. Theirs is learned, differentiable and approximate, which is what makes it usable for planning and also what makes its mistakes invisible. Ours is exact, because it is the generator run forwards. Their scoring is step-level precision, recall and F1 against an expert sequence, together with edge-level IoU; Section 7.1 shows why that family of measures is unsafe in this task, since a model that finds a *shorter* correct sequence is marked wrong by it. That is a finding about scoring in this setting rather than a criticism of their results.

The general precedent for a proposer paired with a verifier is older than any of this work: a network that proposes and a search that checks is the structure behind AlphaGo (Silver et al., 2016), and the reason the combination is stronger than either half.

3.3 COMPUTATIONAL ORIGAMI: THE THEORY UNDER THE ACTION SPACE

The benchmark’s central claim, that geometry does not determine a folded state, is Bern and Hayes’s: deciding the overlap order of a flat folding is NP hard even given a valid mountain-valley assignment (Bern & Hayes, 1996). The action space is Arkin et al.’s: simple folds, in the one-layer, some-layers and all-layers variants, with finite or infinite fold lines (Arkin et al., 2004). Their complexity has been mapped since, including simple foldability’s hardness (Akitaya et al., 2017) and the infinite all-layers case (Akitaya et al., 2019). The solver whose constraint vocabulary we borrow for terminal states is Flat-Folder (Akitaya et al., 2024).

Two results shape what the task actually is. Demaine et al. (2004) showed that any well-behaved folded state is reachable by *some* continuous motion; the question is therefore never whether a path exists, but whether one exists through a finite number of the discrete operations a hand or a machine can perform. And counting foldings has no closed form even for a one-dimensional strip, a problem open since Lucas attributed it to Lemoine in 1891 (Lucas, 1891; OEIS Foundation Inc., 2026); this is the answer to “why not enumerate the states”, and it is a fact about the problem rather than about any implementation.

Finally, the problem itself is not new: generating folding sequences from crease patterns was named and attacked symbolically by Akitaya et al. (2013), by the same group whose tools this field now depends on. What is new is neither the problem nor the theory, but the observation that this problem supplies an exact verifier and can therefore serve as an evaluation substrate for models that reason with tools.

3.4 THE STRUCTURAL GAP

The two closest benchmarks are worth setting beside this one directly, because a reader who knows them will ask what is left to do.

Table 1: The two closest benchmarks beside this one. The informative rows are the last three.

	GAMIBENCH	ORIGAMISPACE	CP2SEQ (OURS)
Model produces	A choice among rendered options	A predicted pattern, a relation, or crease-pattern code	A sequence of folds, one step at a time
Judged by	Agreement with the stored answer	Similarity to a stored target, or compilation	Execution of each step, then replay of the whole
Judge is	A stored key	A comparison against a reference artifact	The fold engine itself, run forwards
Judge can be wrong	Not applicable; no execution	Where similarity stands in for correctness	Only if the model of paper is wrong, which replay cannot detect (Sec. 5.8)
Wrong answers	186 authored impossible patterns	Not a designed axis	Unlimited: an illegal fold is produced by asking for one
Ground truth	Annotated per instance	Annotated per instance	Constructed; recorded as the fold happens
Non-unique answers	One key per question	Compared against one reference	Replayed and compared up to a declared symmetry group (Sec. 7.2)
Scores search effort	No	No	Yes: queries to solution, over all attempts (Sec. 7.3)

Read across the bottom three rows rather than the top. GamiBench’s impossible patterns are the closest the literature comes to forcing a judgement of feasibility rather than a recognition of shape,

and they are authored, which is why there are 186 of them; here the same object is a by-product of generation and there is no limit on it. OrigamiSpace’s instrumentation per instance is richer than ours and is the reason 350 instances sustain a benchmark paper; what it does not have is a judge that can rule on a step the authors never anticipated. Neither ranks among the several valid folded states of one crease pattern, because both compare against a stored answer, and a stored answer cannot represent a set.

None of this is a defect in either benchmark. Both were built to measure whether a model understands folded geometry, and they measure it. The claim here is narrower: that measuring whether a model can *search* for a folding sequence requires a judge that executes, and that origami supplies one.

Collecting the metrics used across this literature makes one absence visible. Every metric in use compares a produced artifact against a target: geometric or semantic similarity to a reference, precision and recall against an expert sequence, IoU over affected edges, compilation success, human preference. **None ranks among the several valid folded states of a single crease pattern, and none scores an agent by how much verification it required.** The absence is structural rather than accidental, because computing any of those metrics requires a target to compare against, and the interesting question here has more than one right answer.

4 PROBLEM FORMULATION

4.1 FLAT-FOLDED STATES

A flat-folded state is an isometric map of the sheet into the plane together with a layer ordering. The map is fixed by the crease pattern; the ordering is not. A single crease pattern therefore admits many folded states that are geometrically identical and distinct as embeddings, and determining the ordering is NP hard even when a valid mountain-valley assignment is given (Bern & Hayes, 1996).

This distinction is the reason the task cannot be reduced to geometry. Over a folding sequence the crease graph’s topology never changes: the same vertices, edges and faces exist at step one and at step nineteen. What changes is everything else. Vertex coordinates are reflected at every step. The overlap structure, which faces lie above which, changes at every step and follows from the geometry. The layer ordering changes at every step and does *not* follow from the geometry; it is chosen, and it is where the decisions live. The number of layers accumulates, which is the formal content of “it gets harder as it goes”.

4.2 THE ACTION SPACE

Simple folds come in named variants, and any benchmark in this area has to say which it uses, because the complexity results differ between them.

Table 2: Named variants of the simple fold (Arkin et al., 2004).

MODEL	DEFINITION
One-layer simple fold	Rotate a single layer $\pm 180^\circ$ about a line
Some-layers simple fold	Rotate a contiguous run of layers
All-layers simple fold	Rotate every layer the line crosses, as sheet metal bends
Finite vs infinite line	Whether the fold line is a full line or a segment

CP2Seq instantiates the all-layers and some-layers tiers. The all-layers tier is the simpler object and it buys three properties at once, the third of which is rarely named. Because the whole stack moves as a rigid body, no two connected pieces of paper ever move relative to each other except at the fold line itself, where paper is allowed to bend. Self-intersection is impossible, the layer ordering is forced rather than chosen, and **tearing is impossible by construction.**

Moving only some layers brings all three back, and tearing becomes the binding constraint. If a moving face is joined to a stationary face along a crease, and that crease is not on the fold line, the

fold would rip the sheet. This is decidable only in the coordinates of the original square, because adjacency is a fact about the original sheet that survives any amount of folding and cannot be recovered from the current positions of the polygons alone.

An example makes the tier difference concrete, and it needs only two folds. Fold the right half of a square over to the left across the vertical midline. The result is a two-layer stack whose top layer is joined to the bottom layer along the entire crease at the midline. Now select the top layer alone. Folding its upper half down across the horizontal midline would tear the sheet, because the moving piece is joined to stationary paper along a vertical crease segment that is perpendicular to the fold line. Folding its left quarter across a vertical line does not tear, because the moving piece’s only join to stationary paper lies on the fold line itself. Same state, same layer selection, opposite verdicts, and the only difference is the orientation of the line.

4.3 THE TASK

Given a crease pattern, produce a sequence of simple folds that reproduces it. Generating an instance costs one forward pass of the engine; solving it is NP hard. That asymmetry is the paper’s foundation.

Caveat. One claim must not be made, and it was drafted wrongly here once. It is tempting to say that legal moves are rare and that this is what makes the task hard. Enumeration refutes it: legal partial folds *grow* with depth, from 22 at two layers to 332 at thirty-eight, six times the 56 all-layers folds available at the same state. What falls is the hit rate of uniform random proposal, from 11.9% to 3.8% by seventeen layers, because the space being sampled grows faster than the legal set inside it. That is a fact about a sampler, not about origami, and conflating the two would put a false statement about branching factor into the paper.

5 CP2SEQ: THE DATASET

5.1 WHY THE CORPUS IS SYNTHESISED

Every sample is built rather than collected, and the reason is that the construction supplies the label. A folding sequence is not something that can be read off a finished crease pattern by inspection, which is the whole premise of the task; so a corpus of annotated patterns would require either a solver, which is the thing being benchmarked, or a human expert, which does not scale and introduces an error rate nobody can measure. Folding forward and recording avoids both. The sequence is not inferred after the fact, it is what happened, and the pattern is what the folding left behind.

This also means difficulty is a dial rather than an observation. A collected corpus has whatever distribution of depth and entanglement its source happened to have. A generated one is stratified on purpose, and the empty regions of that stratification are visible rather than hidden (Section 5.7).

5.2 GENERATION, STEP BY STEP

The generator starts from a unit square and applies randomly chosen legal simple folds, recording each one, until it reaches a target depth. It then unfolds and reads off the crease pattern.

```

GENERATE(seed, depth_target, tier):
    rng    <- seeded RNG from 'seed'           # replay depends only on this
    state <- unit square, one layer
    seq    <- []

    while len(seq) < depth_target:
        cands <- ENUMERATE_LEGAL_FOLDS(state, tier) # every fold the engine accepts
        if cands is empty:
            return FAILED                          # dead end: discard, never repair
        fold  <- rng.choice(cands)
        state <- APPLY(state, fold)                # the APPLY the verifier uses
        seq.append(fold: line, direction, layers, creases made, coupling)

    cp      <- UNFOLD(state)                      # creases accumulated over the whole sequence

```

```

cp      <- PLANARISE(cp)      # split edges at crossings; canonical form
frames <- [state after each step] # retained as steps.fold

assert REPLAY(cp, seq) == state # sample rejected if this fails
return Sample(cp, seq, frames, metrics(cp, seq))

```

Four properties of this procedure matter for the benchmark and each is a deliberate choice.

The generator never searches. It picks uniformly among legal folds and never backtracks. A dead end discards the sample rather than repairing it. This keeps generation one forward pass, and it is also why the recorded sequence carries no claim to being minimal (Section 7.1).

ENUMERATE_LEGAL_FOLDS and APPLY are the same functions the environment exposes to the model. The generator is not a privileged path through different code; it is the environment driven by a random policy. A sample is therefore reachable by definition, using exactly the action space the model is given.

Planarisation is canonical. A crease pattern is stored with every edge split at every crossing, so two patterns describing the same geometry have the same edge set and can be compared as multisets without a tolerance.

Every sample is replayed before it is kept. The assertion is not a test that runs sometimes; a sample that does not reproduce its own state is discarded at generation time.

5.3 REPRODUCIBILITY AS A SHIPPING FORMAT

Given its seed and the generator version, a sample regenerates byte-identically. The corpus is distributed as a generator plus a manifest, not as a data archive: the manifest lists each sample’s seed, tier, target depth and a hash of the resulting crease pattern, and rebuilding is verification. A reader who wants ten thousand samples instead of six hundred, or a stratification we did not choose, runs the generator rather than asking us for a larger download.

5.4 DIFFICULTY, STRATIFIED RATHER THAN SAMPLED

Difficulty is set along two axes rather than measured after the fact. **Fold depth** is the primary axis: the number of folds in the recorded sequence. It is the axis the task’s hardness is defined on, because step k is constrained by steps 1 through $k - 1$. **Coupling** is the second: the number of creases a single fold creates, which is to say how many layers the fold line cuts. It is free to compute at generation time, since one fold cuts every layer it crosses and each cut becomes one crease in the unfolded square. Coupling is the measurable proxy for non-local dependency: a high-coupling fold writes creases into many layers at once, so a later fold cannot be reasoned about locally. The two axes are not independent, and the dependence is itself a finding (Section 5.7).

5.5 WHAT THE CORPUS CONTAINS

The release corpus holds 600 samples in five batches, divided into two splits that differ in what is *known* about each sample. The distinction matters when reading numbers elsewhere in this paper. Every sample is correct by construction, because it was produced by folding forward and recording. The *verified* split additionally carries a solver verdict, so it is the only split on which statements about solvability or about shorter sequences can be made; the shorter-sequence rate of Section 7.1 is measured there and nowhere else. The *generated* split has no verdict. The corpus-level checks of Section 5.8 run over all 600 samples, because replay and crease comparison need no solver.

Table 3: The release corpus. Splits differ by what is known, not by how samples were made. Crease and layer figures are medians, with the maximum in brackets.

BATCH	TIER	SPLIT	n	FOLDS	CREASES	LAYERS
all-layers	all-layers	generated	400	4–19	83 (24448)	58 (16384)
some-verified-d3	some-layers	verified	50	3		
some-verified-d4	some-layers	verified	50	4	10.5 (75)	9 (42)
some-generated-d5	some-layers	generated	50	5		
some-generated-d6	some-layers	generated	50	6		
Verified split			100	3–4		
Generated split			500	4–19		
Total			600	3–19	33 (24448)	23.5 (16384)

Across the whole corpus the median crease count is 33 and the median layer count 23.5, with the deep tail of the all-layers tier reaching tens of thousands of both. The some-layers tier is shallower by design and its medians are an order of magnitude smaller, which is why the two tiers are never pooled when results are reported.

5.6 WHAT A SAMPLE CARRIES

Table 4: The files shipped with each CP2Seq sample. Only the first two, and only the final frame of the second, are shown to the model.

FILE	CONTENTS
<code>cp.fold</code>	The crease pattern, planarised. The input
<code>steps.fold</code>	The pattern plus the folded state after each step, as one multi-frame file. The final frame is the other input ; the intermediate frames are held back
<code>seq.json</code>	Every fold: line, direction, selected layers, creases made, coupling. The ground truth , never shown to the model
<code>meta.json</code>	Difficulty metrics, degeneracy flags, seed and generator version

DRAFT IMAGE — PLACEHOLDER, NOT FINAL

This figure is a draft. It will be replaced with a human-authored figure before submission. Drafts are generated with OpenAI gpt-image-2 via `paper/figures/gen_figures.sh` and used only as tracing references; no generated image carries data or geometry.

Figure 2 placeholder: one dataset sample end to end. Left, the crease pattern; centre, the final folded state as X-ray and exploded views; right, the withheld ground-truth sequence, one diagram per fold.

Figure 1: [DRAFT IMAGE] One CP2Seq sample. The crease pattern and the final folded state are given to the model; the fold sequence on the right is the ground truth and is never shown.

5.7 TWO LIMITS THAT BELONG IN THE PAPER RATHER THAN AN APPENDIX

Caveat. The first is an empty cell. Long sequences at low coupling are almost unreachable, yielding 2hits in 16,000 attempts, because every all-layers fold thickens the stack, so a long sequence cannot keep cutting few layers. Real folders reach that corner by **pre-creasing**, an operation this action space excludes. The longer a real model runs, the more of it sits outside what the benchmark can express.

Caveat. The second is depth. The some-layers tier stops at six folds in this release, and the difficulty the benchmark is about lives past the depth a search can reach. Restoring the deeper tier is a matter of generator configuration and wall-clock, not redesign.

5.8 VERIFYING THE CORPUS ITSELF

Two checks run over every sample, written independently of each other. The first replays each recorded sequence and confirms it reproduces the pattern stored beside it. The second, written without reference to the first, compares maximal creased intervals as multisets with no tolerance in the verdict, after clustering edges into lines so that subdivision differences cannot register as disagreements. Both pass on all 600 samples of the release corpus.

Caveat. Both share the fold engine with the generator, which bounds what they can establish. They cannot catch a wrong model of paper: if the engine is wrong about tearing, replay is wrong in the same way and agrees with itself. What they do catch is drift between what was folded and what was recorded, which is the likelier failure and the one that silently corrupts ground truth. An independent check requires a third-party solver and is not claimed here.

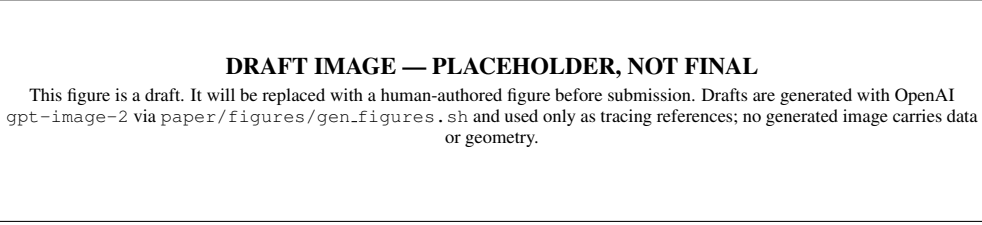


Figure 3 placeholder: the difficulty grid. Fold depth against coupling, shaded by sample count, with the long-sequence low-coupling cell visibly empty.

Figure 2: [DRAFT IMAGE] The difficulty grid. The empty corner is long sequences at low coupling, which this action space cannot reach because it excludes pre-creasing.

6 THE ENVIRONMENT, WHICH IS ALSO THE VERIFIER

6.1 ONE OBJECT, TWO NAMES

In this benchmark the environment and the verifier are the same object. Stepping it is the same act as asking for a verdict: the model submits a candidate fold, and the reply is either the resulting state or a refusal. There is no separate validity oracle, no second artifact that could disagree with the first, and no learned component in the loop.

6.2 THE DIVISION OF LABOUR

Table 5: Who does what. The environment does not search.

COMPONENT	RESPONSIBILITY
Environment / verifier	State update and legality. It does not search and does not choose
Model	Proposes the next fold, decides where to go, recognises when to backtrack

This division is what makes the benchmark measure something. The obvious objection to any tool-augmented setup is that the tool is doing the work, and here the answer is structural rather than rhetorical: deciding global flat-foldability is NP hard, so the environment cannot search its way to an answer even in principle. It can only say whether one proposed step is something paper could do. Compressing an exponential search into a feasible number of queries is exactly what the environment cannot supply and exactly what the model is being measured on. The referee is not a player.

6.3 REFUSALS ARE NAMED

An illegal fold returns a structured refusal with a name, not a boolean.

The names are the design. A model told `would-tear` can act on the information; a model told “illegal” cannot. This also turns error analysis into a taxonomy that exists before the experiment: every rejection arrives already labelled with the class it violated, so a reject is a diagnosis rather than a dead end.

Caveat. There is deliberately no self-intersection refusal, and there cannot be one in this action space. A some-layers fold may only move a contiguous run of layers taken from the top or the bottom of the stack, and such a move cannot drive paper through the layers it left behind. Lifting the top two layers and folding them across is something a hand can do; folding them *underneath* the stack is not a fold but a slit. Restricting selection to a run at one extremity makes the illegal case unrepresentable rather than undetected. The four constraint classes that Flat-Folder checks (taco-taco, taco-tortilla, tortilla-tortilla, transitivity) (Akitaya et al., 2024; Flat-Folder) remain the right vocabulary for global terminal states; they are not what a single legal step needs checking against here.

Table 6: The refusal taxonomy. A rejection arrives already labelled with the class it violated.

REFUSAL	MEANING
would-tear	The moving layers are joined to stationary paper away from the fold line, so the sheet would come apart. The binding constraint of the some-layers tier
no-crease	The fold moves layers without creasing anything, tearing them free of the sheet rather than folding them
nothing-to-move	The fold line misses the selected layers entirely
direction-impossible	A run taken from the bottom of the stack was asked to fold over, or the reverse
bad-line	The fold line is malformed, or names no side of itself

6.4 WHAT THE MODEL OBSERVES

On success the environment returns the new state together with two views: a top-down X-ray of the stack, and an exploded view of the layers. Every intermediate state in this tier is flat, so a second camera angle would show the same silhouette rotated and would carry no new information; what carries information is layer structure, which is why the second view is an explosion rather than a rotation. Positions are reported as coordinates on the original sheet together with an integer layer index. There is no continuous vertical coordinate at this tier, and a rendering that looks three-dimensional is not evidence of one; thickness belongs to a different problem and a different paper.

6.5 ERROR VERBOSITY IS AN EXPERIMENTAL VARIABLE

The richer a refusal, the more of the reasoning the environment performs rather than the model. At the limit, a message that names the fix has solved the step. Verbosity is therefore fixed before the first run and reported with the results; otherwise the tool-augmented condition is not reproducible and its comparison against the verifier-only condition measures message design rather than reasoning.

7 SCORING

7.1 WHY THE OBVIOUS PROTOCOL IS WRONG

The recorded sequence is *a* solution, not *the* solution. The generator folds; it does not search, so nothing makes its sequence minimal, and no minimal-length label is shipped because producing one would require a search this benchmark deliberately does not run. Measured on the verified split, 2of 100samples admit a sequence one fold shorter than the one that built them.

The consequence is sharp. **Any metric that compares a proposal step by step against the recorded sequence marks a better answer wrong.** A model that finds a shorter correct solution is penalised for it. This disqualifies edit distance, and it disqualifies step-level precision, recall and F1 against an expert sequence for this task. Step count is not a correctness signal in either direction.

7.2 WHAT WE DO INSTEAD

Proposals are scored by replaying them through the engine and comparing the resulting states. Both sides are replayed: the model’s sequence and the reference sequence both pass through the same engine, and a stored frame is never the object compared against. That last point is a consequence of the file format rather than a preference. Stored frames record where each face sits in the current plane and not which piece of the original sheet it is, and deep in a stack many layers are congruent triangles, so two states differing by a swap of two such layers cannot be distinguished from stored frames alone. A replayed state carries original-sheet coordinates for free, because the engine needs them to decide tearing at all.

Two levels of equality are reported.

Level 1, crease sets. An exact multiset comparison of maximal creased intervals, equal rather than overlapping, with no tolerance in the verdict. This level is genuinely tolerance-free: edges are clustered into lines first, so subdivision differences cannot register as disagreements, and what remains is a combinatorial comparison.

Level 2, folded states up to a declared group of transforms. Because one crease pattern admits many valid terminal states, exact equality against the one stored state would mark correct answers wrong. The answer is an equivalence: declare in advance which differences do not count, then demand equality inside that.

The transforms are quotiented out of every result reported in this paper. The declared group is the full plane isometry group: **arbitrary translation, arbitrary rotation, and reflection**. One single isometry must carry every layer of the candidate onto its reference counterpart; a match is not accepted by transforming each layer independently.

Each transform is in the group for a concrete reason rather than for generality. *Translation*, because a fold can carry paper off the original square and which half of the sheet travels decides where the stack lands; a folded model is the same model wherever it sits on the table. *Rotation*, because the same object produced by a sequence that worked around the square in a different order arrives rotated; rotations preserve both stack order and face parity, so they need no further correction. *Reflection, with two corrections*, because a reflection represents turning the model over, and turning a model over mirrors the coordinates, **reverses the stack and flips every face’s parity** at once. All three are applied together. Applying a reflection to coordinates alone would compare a model against its mirror image and silently accept wrong answers, which is the most dangerous failure a comparator of this kind can have.

Floating point is unavoidable once transforms are admitted, and this paper states its tolerance rather than claiming not to have one. A rotation by an angle that is not a multiple of a right angle produces irrational coordinates, so candidate and reference vertices agree only to within floating-point representation, and comparing them for bitwise equality would reject correct answers. The comparator matches polygons under a tolerance of 2×10^{-6} in the plane, with degenerate-vertex cleanup at 10^{-7} and a collinearity threshold of 10^{-9} . These values are reported in every result record, so a reader can see what was used rather than infer it.

Caveat. This qualifies a claim made elsewhere. Level 1 crease-set comparison is genuinely tolerance-free, since the verdict involves no floating-point threshold. Level 2 is not, and cannot be, because it admits rotations. The precise formulation is that the *verifier* is exact, since legality is decided combinatorially by the engine, while *terminal-state equality under transforms* carries a stated numerical tolerance.

Caveat. The layer-order variants that a flat-foldability solver reports as equally valid are deliberately *not* in the group. Those are not symmetries of one state; they are different states reachable from the same crease pattern, and deciding which orderings are valid requires a solver. Folding that into an equality test would hide an external dependency inside what reads as arithmetic. If that equivalence is wanted, it belongs in a separate check that owns the solver.

7.3 METRICS

The primary metric is **solve rate at a fixed query budget**, reported per difficulty stratum and never pooled, under both levels of equality. A proposal is solved if replaying it reproduces the target. Exhausting the budget is a timeout, which is reported separately from a wrong answer because the two mean different things.

The secondary metric is **queries to solution**, reported over all attempts including those that time out. Conditioning on the solved attempts hides precisely the long-tail blowup that makes the task interesting. This metric is what answers the “the simulator did all the work” objection quantitatively: two models sharing one exact verifier can be ranked by how little of it they needed.

7.4 VALIDATING THE PROTOCOL

A verifier paper that does not test its own verifier is asking to be taken on trust, so the comparator ships with a negative control: a deliberately corrupted state, with two layers swapped, must be rejected.

The control earned its place. On its first full run, 23 of 600 corrupted states were **accepted**. The cause was not a tolerance. The comparison used current-plane geometry and parity alone, and deep in a stack many layers are congruent triangles, so two layers occupying the same region but originating from different parts of the sheet were interchangeable. They are not interchangeable: which one lies underneath is a real difference between two folded states, and precisely the difference Level 2 exists to catch. With face identity on the original sheet restored to the comparison, the full corpus reports 600 states equal, all under the identity symmetry, and 600 of 600 corrupted states rejected. Every verdict additionally reports whether face identity was available, so a weakened comparison is visible in the output rather than assumed away.

8 OBJECTIONS, ANSWERED HERE RATHER THAN IN REBUTTAL

“The simulator does all the work.” It cannot: deciding flat-foldability is NP hard, and the environment only rules on single steps. Section 6.2 states the division of labour and Section 7.3 measures it.

“Origami is a toy problem.” Flat origami is Turing complete (Flat Origami is Turing Complete), so the domain is not expressively impoverished. **Caveat.** This is a defensive citation and nothing more. It shows that the domain is rich enough to be interesting; it does not show that anything learned here transfers, and the causal chain from “origami is Turing complete” to “training on origami yields physical understanding” is broken in the middle. Rule 110 is Turing complete too.

“Why not enumerate the valid states?” There is no closed form for the number of foldings even of a one-dimensional strip, a problem open since 1891 (Lucas, 1891; OEIS Foundation Inc., 2026; Koehler, 1968), and state counts for real patterns reach astronomical magnitudes. The hardness is the problem’s, not an implementation’s.

“Synthetic data is a shortcut.” The comparable published corpora are largely produced by their authors’ own symbolic simulators; synthesis is the field’s normal practice. Here it is also what makes difficulty controllable and ground truth constructed rather than annotated, and what allows the corpus to ship as a seed rather than as a file.

“Your corpus is not real origami.” Correct, and measured: 89.3% of the 366 real crease patterns in Flat-Folder’s `examples/instagram/` corpus (Flat-Folder) provably lie outside all-layers simple folding. This is stated as a scope boundary in Section 2 and Section 12 rather than left for a reviewer to discover.

“The model may have memorised the pattern.” The no-tools arm runs on anonymized geometry with file-names and metadata stripped, for exactly this reason: if a model can read a pattern’s name it recalls rather than reasons, and then both arms measure recall and the comparison measures nothing. See Section 10.

9 EXPERIMENTAL SETUP

9.1 THE PIPELINE

One episode is one sample attempted by one model under one tier of assistance. The loop is the same for every arm and every model, and the only thing that varies between arms is what the model is allowed to see and call.

```
EPISODE(sample, model, tier, budget):
    cp, final <- LOAD(sample)      # cp.fold and the LAST frame of steps.fold
                                   # intermediate frames are never exposed
    state <- unit square, one layer
    seq <- []
    while len(seq) < budget:
        obs <- RENDER(state, tier)  # views withheld at the no-vision tier
        action <- model(cp, final, obs, history, tier)
        if action is SUBMIT: break
        verdict <- ENVIRONMENT.step(state, action)  # the verifier IS this call
        if verdict is REFUSAL:
            history.append(refusal with its named reason)  # state unchanged
        else:
            state <- verdict.state
            seq.append(action)
            history.append(obs)
    return SCORE(replay(cp, seq), replay(cp, reference_seq))
```

Three things about this loop decide what the benchmark measures.

The environment is the only thing that changes state. A model cannot assert that a fold happened. It proposes, and the environment either performs the fold or refuses it, so the state the model reasons about is always a state paper could be in.

A refusal costs a query and does not advance the episode. This is what makes queries-to-solution meaningful: a model that proposes carelessly pays for it in budget, and a model that reasons before proposing is rewarded in the secondary metric even when both eventually solve the sample.

The reference sequence is never read during the episode. It enters only at scoring time, and then only as something to replay, never as something to compare against step by step (Section 7.1).

9.2 THE TOOLS THE MODEL IS GIVEN

The tool belt is the experimental variable. Every tool is a read or a step against the same engine; none of them searches, and none of them ranks candidates.

DRAFT IMAGE — PLACEHOLDER, NOT FINAL

This figure is a draft. It will be replaced with a human-authored figure before submission. Drafts are generated with OpenAI gpt-image-2 via `paper/figures/gen_figures.sh` and used only as tracing references; no generated image carries data or geometry.

Figure 4 placeholder: the evaluation pipeline. Inputs on the left; the propose-verify-observe loop in the centre with the environment and verifier drawn as one box and refusals looping back without advancing; replay and scoring on the right, with the reference sequence outside the loop.

Figure 3: **[DRAFT IMAGE]** One episode. The environment and the verifier are the same object, so stepping is the same act as asking for a verdict. The reference sequence enters only at scoring.

Table 7: The tool belt. Tiers are defined by which of these are available.

TOOL	WHAT IT RETURNS	WHY IT EXISTS
<code>apply_fold</code>	The new state, or a named refusal	The action. This call is the verification
<code>get_views</code>	Top-down X-ray of the stack, exploded layer view	The visual channel; withheld at the no-vision tier
<code>get_state</code>	Face positions in original-sheet coordinates plus integer layer index	The symbolic channel, so a model is never forced to read geometry off a raster
<code>list_legal_folds</code>	Every fold the engine would currently accept	The filtering tier. It prunes; it does not choose
<code>undo</code>	The previous state	Backtracking without spending the episode

`list_legal_folds` is the tool that needs justifying, because it is doing the most work. It enumerates candidate actions and returns those that are legal, which is a large prune performed by the simulator rather than by the model. That is precisely why the tier structure exists and why the deterministic baseline of Section 9.4 is reported: if exhaustive search over the enumerated set solves the corpus outright, then the filtered arm is not measuring origami reasoning, and the benchmark has to say so rather than let a headline number imply otherwise.

DRAFT IMAGE — PLACEHOLDER, NOT FINAL

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Figure 5 placeholder: a refusal as the model receives it. The proposed fold line over the current state, the named reason would-tear, and the offending crease segment highlighted; beside it the same state with a legal fold line.

Figure 4: **[DRAFT IMAGE]** A refusal is a diagnosis, not a dead end. The same state and the same layer selection give opposite verdicts depending only on the orientation of the fold line.

9.3 TIERS, MODELS AND PROTOCOL

Tiers. Four levels of assistance, from a full tool belt down to none; the table is in Section 10. The same model, the same samples and the same query budget across tiers, so that the only difference between two tiers is what was withheld.

Models. [TO FILL from the aggregation: the models actually run, their versions, and the reasoning-effort and image-history settings used. Held fixed across tiers, because comparing across models and across tiers at once would confound the two.]

Protocol. Query budget, number of repeats and seeds are fixed before the first run and reported with the results. Median and spread across repeats are reported, never a single run, because sampling is not deterministic and one run is not a measurement.

Transforms are applied in every reported result. Every solve rate in Section 11 is computed after quotienting out the plane isometry group of Section 7.2: translation, rotation and reflection, with reflections reversing the stack and flipping face parity. A candidate is counted correct if one single isometry carries every layer onto its reference counterpart. This is not a post-hoc leniency applied to borderline cases; it is part of the definition of a correct answer, applied identically to every arm, every model and every baseline. Turning it off would mark correct answers wrong, because where a folded model lands on the table is an accident of which half of the sheet travelled.

Matching under transforms requires a floating-point tolerance, and it is stated. Admitting arbitrary rotation means candidate and reference coordinates agree only to within floating-point representation, so the comparator matches polygons within 2×10^{-6} , cleans degenerate vertices at 10^{-7} and treats vertices as collinear below 10^{-9} . Every result record carries the tolerance it was scored under, so the value is visible in the released runs rather than buried in a constant. The engine’s legality decisions are combinatorial and involve no tolerance; the numerical threshold applies only to terminal-state equality under transforms.

Error verbosity is itself a variable. The richer a refusal, the more of the reasoning the environment performs rather than the model; at the limit a message that names the fix has solved the step. Verbosity is fixed before the first run and reported, otherwise the comparison between tiers measures message design rather than reasoning.

9.4 BASELINES

Three, none of them a language model, so the models are ranked against something rather than only against each other.

Table 8: The baseline ladder. [TO RUN]

BASELINE	WHAT IT ESTABLISHES
Uniform random over enumerated legal actions	The floor. What the tool alone achieves with no reasoning at all
Deterministic breadth-first over the same action set	How much of the task the enumerator already solves, and the depth at which exhaustive search stops being affordable
Symbolic crease-pattern-to-sequence solver (Creasy)	The closest thing to prior art on the task itself

The breadth-first baseline is the one that answers the “the simulator did all the work” objection of Section 8 numerically rather than structurally, which is the stronger form of that answer.

10 PROGRESSIVE TIERS OF TOOL ASSISTANCE

[TO RUN]

The model is run at four levels of assistance, each removing one kind of help from the level above. These are reported as tiers rather than as ablations, for the reason given in Appendix ??: a full ablation study was planned and has not been run, and calling a partial sweep an ablation would overclaim.

Table 9: The tiers of assistance. Same model, same patterns, same query budget across tiers.

TIER	TOOLS AVAILABLE	WHAT IT ISOLATES
Full tool belt	Environment, rendered views, any auxiliary tools	Upper bound
No vision	Same, rendered views withheld	The contribution of visual feed-back
Verifier only	Pass/fail, no filtering	The contribution of filtering on top of raw verification
No tools	None, anonymized geometry	Memorization control

Same model, same patterns, same query budget across tiers. If the no-vision or no-tools tier performs nearly as well as the full tier, **that is a finding rather than a failed experiment**: it means the gain is not where it was assumed to be. [TO RUN]

11 RESULTS

All numbers below come from an aggregation over every saved run: 1,211 attempts across 47 runs and 256 distinct samples. Every attempt is counted once, and an attempt with no recorded `solved` field is counted as unsolved.

11.1 HEADLINE

No language model solved a single sample beyond the easy stratum. Across every model arm, mid is 0 of 70 and hard is 0 of 27. On the easy stratum a deterministic breadth-first search over exactly the action set the models are given solves 54.7%, against 20.4% for the best-covered model at the same tier. **Blind search beats every model tested, by a factor of about 2.7 on the only stratum where anything succeeds at all.**

Table 10: Solve rate per stratum, all saved runs. Denominators are attempts, not distinct samples. Rows with fewer than ten attempts support no rate and are shown only so that reporting is not selective.

ARM	TIER	EASY	MID	HARD	OVERALL	CALLS
deterministic-bfs	legal-folds	366/669 (54.7%)	1/40 (2.5%)	0/16	367/725 (50.6%)	5
gpt-5.6-luna	legal-folds	59/289 (20.4%)	0/66	0/25	59/380 (15.5%)	23.5
gpt-5.6-luna	no tools	4/65 (6.2%)	0/1	0/1	4/67 (6.0%)	29
claude-sonnet-5	legal-folds	0/5	–	–	0/5	10
gpt-5.6-sol	no tools	0/7	–	–	0/7	20
gpt-5.6-terra	no tools	0/1	–	–	0/1	35
gpt-6-astra	no tools	1/1	–	–	1/1	5
codex-cli-chatgpt	none	0/1	–	–	0/1	2
<i>errored, model unrecorded</i>	–	0/20	0/3	0/1	0/24	–

Caveat: read the denominators before the percentages. Only `gpt-5.6-luna` and the baseline have coverage worth a rate. `gpt-6-astra` shows 1/1; that is one attempt and is not a 100% solve rate. Four arms have between one and seven attempts each. These rows are present because omitting them would be selective reporting, not because they support a comparison.

11.2 THE BASELINE READING

`deterministic-bfs` solves 367 of 725 attempts: 366/669 easy, 1/40 mid, 0/16 hard. Median states expanded is 65 on easy, 1051.5 on mid and 532 on hard. 358 of its 725 attempts ended in timeout rather than exhaustion, so 50.6% is a lower bound under the baseline’s time budget rather than a ceiling on exhaustive search.

The enumerator evaluates roughly 2,010 candidate actions per state and returns a mean of 3.51 that are legal and stay inside the target crease pattern, over 73,750 enumerations. A branching factor near 3.5 makes an easy

sample of five folds a tree of a few hundred states, and the median of 65 states expanded on easy confirms it. **The easy stratum is therefore not measuring origami reasoning.** It measures whether a model can avoid losing to breadth-first search on a tree its own tool has already pruned roughly 570-fold, and every model tested loses.

We state this rather than report the models’ easy-tier rate as an achievement. It is also the answer to the “the simulator does all the work” objection of Section 8, and the answer is uncomfortable: on the easy stratum the simulator does most of the work. The benchmark’s value here is that it measures that rather than hiding it. On mid and hard the objection dissolves, because search solves 1 of 56 and the models solve 0 of 97.

11.3 WHAT THE TOOLS CONTRIBUTE

The one within-model tier comparison the data supports is gpt-5.6-luna on easy: 59/289 (20.4%) with the filtering tool against 4/65 (6.2%) with no tools, so filtering is worth roughly 3.3× on that stratum and nothing on mid or hard, where both tiers are zero.

Caveat. The no-vision and verifier-only tiers of Section 10 were not run. The four-tier ladder this paper describes is not yet instantiated; only its two endpoints exist.

11.4 FAILURE TAXONOMY

Table 11: Termination reasons for gpt-5.6-luna, 447 attempts.

TERMINATION	<i>n</i>	SHARE
state_cycling	215	48.1%
finished	167	37.4%
turn_budget	47	10.5%
repetition_detected	18	4.0%

This is the most informative result in the section, and it is not a solve rate. A majority of attempts, 233 of 447 counting cycling and repetition together, end with the model revisiting a state it has already produced rather than exhausting its budget. The model is not running out of room to search; it is failing to notice it has been somewhere before. That is a specific, diagnosable deficit in maintaining state across a multi-step spatial task, and it is the kind of finding a benchmark with an executing verifier can produce and a multiple-choice benchmark cannot. The baseline never cycles, because breadth-first search over a visited set cannot; it times out instead, 358 times.

11.5 WHERE DIFFICULTY LIVES

Depth separates the arms completely. Easy is the only stratum where anything succeeds; mid yields a single solve, by the baseline, at a median of 1051.5 states expanded against 65 on easy; hard yields none from any arm. [TO RUN: solve rate against coupling, to test whether the second stratification axis of Section 5.4 predicts difficulty as intended. Only depth is evidenced so far.]

11.6 THREATS TO THESE RESULTS

Every model attempt ran at low reasoning effort. All 461 language-model attempts with a recorded setting used reasoning_effort: low; 25 have none recorded. This is the single largest caveat on every claim in this section. “Frontier models fail this task” is not supported by these runs. What is supported is that frontier models at low reasoning effort fail this task and lose to breadth-first search where search works at all. Re-running the best-covered arm at high effort is the most valuable remaining experiment, and until it exists the headline carries the qualifier.

Coverage is severely imbalanced. 725 baseline attempts and 447 for gpt-5.6-luna, against between one and seven for every other model. No cross-model claim is made.

24 attempts terminated in error with no model recorded and are counted as unsolved. They are reported as their own row rather than dropped, because dropping failed attempts inflates every rate above them.

Repeats are not yet reported as median and spread. Section 9.3 promises this; the aggregation counts attempts rather than grouping repeats per sample. The protocol requires it before submission.

DRAFT IMAGE — PLACEHOLDER, NOT FINAL

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Figure 6 placeholder: solve rate per difficulty stratum, one line per model, with the deterministic baseline as a distinct reference line crossing above every model line on the easy stratum, and attempt counts printed at each point.

Figure 5: [DRAFT IMAGE] Main result per stratum. The baseline crosses above every model line on easy; that crossing is the most informative feature of the chart and is not smoothed away.

12 LIMITATIONS

Every model attempt ran at low reasoning effort. All 461 language-model attempts with a recorded setting used `reasoning_effort: low`. The results support a claim about models at that setting, not about frontier models in general, and the headline is worded accordingly. Raising the effort on the best-covered arm is the most valuable remaining experiment.

The easy stratum does not measure what the benchmark is for. Deterministic breadth-first search solves 54.7% of it at a median of 65 states expanded, beating every model. The stratum measures whether a model can beat trivial search on a tree its own tool has pruned roughly 570-fold. Claims about reasoning should be read off mid and hard, where every arm including search is at or near zero, and where the benchmark currently discriminates nothing either. **The band in which this benchmark separates models may be narrow or, at this depth range, empty.**

Model coverage is severely imbalanced. 447 attempts for one model and between one and seven for four others. No cross-model comparison is supported.

No training. Every arm is an off-the-shelf model driven by prompting and tool calling; no model is trained or fine-tuned. The contribution is the harness, not a model. This is stated in the introduction as well as here, because a limitation the authors declare is context and one a reviewer discovers is an objection.

Scope of the action space. 89.3% of the 366 real crease patterns in Flat-Folder’s `examples/instagram/` corpus lie outside all-layers simple folding, and pre-creasing, the operation real folders use to reach long sequences at low coupling, is excluded.

The corpus verifiers share an engine with the generator, so they cannot detect a wrong model of paper. They detect drift between what was folded and what was recorded.

Flat states only. No thickness, no material, no mechanics. Thickness changes the geometry outright and belongs to a separate line of work with a verifier several orders of magnitude more expensive.

Level 2’s group is narrower than the truest notion of equality, because valid layer-order variants are excluded for the reason given in Section 7.2. It is therefore strict rather than permissive, which is the safe direction but not a free one.

13 FIGURES

14 REPRODUCIBILITY

The corpus ships as a generator and a manifest; every sample rebuilds byte-identically from its seed. The environment, both corpus verifiers and the state comparator are released with the benchmark. The negative control of Section 7.4 runs as a command, so a reader can confirm that the comparator rejects corrupted stacks rather than taking Section 7.4 on trust.

Table 12: Planned figures and their status.

#	FIGURE	
1	The asymmetry: generation is one forward pass, recovery is NP hard	To draw
2	The tearing pair of Section 4.2. One state, one layer selection, a vertical fold line legal and a horizontal one not	To draw
3	A refusal as the model receives it: view, named reason, per-constraint checklist with locations	To draw
4	The difficulty grid, depth against coupling, with the empty cell visible	To draw
5	Main result per stratum	[TO BE DETERMINED]

15 LICENCE AND AVAILABILITY

The corpus generator, the environment and verifier, both corpus checks, the state comparator and the evaluation harness are released under the **BSD 3-Clause** licence.

Code and data are released on publication; links are omitted here.

The artifacts carry no personal data, no human-subject data and no scraped content. Every sample is synthesised from a seed by folding a square, so no licensing question attaches to the geometry and no attribution is owed to any origami designer. Flat-Folder’s `examples/instagram/` set, the one external corpus this paper measures against, is not redistributed; the scope measurement of Section 12 was computed against it in place and is reported as a number.

16 USE OF LARGE LANGUAGE MODELS

ICLR 2026 requires that any use of a large language model be disclosed, on the principle from the Code of Ethics that all contributions to the research must be acknowledged, and holds the authors responsible for everything in the paper regardless of what assisted in producing it. This section is that disclosure. It is written to be complete rather than minimal, because a disclosure that omits a use is worse than one that reports a use a reader would have forgiven.

Large language models are part of the object of study. This is a benchmark for evaluating language models, so models appear in the work by construction. Every number reported in Section 11 is the output of a frontier multimodal model driven through the harness described in Section 9. The models evaluated, their versions and their settings are listed there. No model was trained or fine-tuned for this paper.

Code. The benchmark’s implementation was written with AI assistance, principally Claude Code, used as a coding assistant throughout. Every design decision, every experiment, and the direction of the work were determined by the human authors, and all generated code was reviewed by them before it was used. The authors are responsible for the correctness of the released artifacts. Two habits in the project exist because of this reliance and are worth naming: the corpus carries two independently written checks over every sample (Section 5.8), and the state comparator ships with a negative control that must reject a deliberately corrupted state (Section 7.4). The second of these caught a real defect, described in that section, in which 23 of 600 corrupted states were initially accepted.

Literature search. The related work of Section 3 was assembled primarily by hand, using Google Scholar and alphaXiv. AI assistance was used partially, to locate candidate papers and to recover bibliographic details. Every citation in this paper was verified against the arXiv or publisher record by a human author; citations that could not be verified were removed rather than softened.

Writing. AI assistance was used in drafting and editing the manuscript. The argument, the claims, the experimental design and the conclusions are the authors’. No text was included that an author had not read and agreed with, and no result, citation or number in this paper was produced by a language model without being checked against the artifact it describes.

Figures. Any figure marked **DRAFT IMAGE** is a placeholder generated with OpenAI `gpt-image-2` and is used only as a compositional reference to be redrawn by hand. Generated images are never used to depict data or geometry: crease patterns, folded states and result charts are rendered from the corpus and from the run outputs. Draft figures will not appear in a submitted version of this paper.

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A REFERENCE NOTES

The bibliography is grouped in `iclr2026_conference.bib` by the role each work plays in the argument rather than alphabetically, so that a co-author can see what is load-bearing. The notes below record what is still outstanding.

Origami benchmarks and systems for language models. Xu et al. (2025) must be cited as a preprint; `papers.md` recorded NeurIPS’25 and the arXiv record carries no venue. Spencer et al. (2025) has code and data released. Moriya et al. (2026) reconstructs folding processes from instructional video and is built on the PurelandFold dataset; code at <https://github.com/maya-moriya/FoldingAgent>. **OrigamiBench** (Agarwal et al., 2026) has a *citation missing*: it appears in our metrics survey as the source of query efficiency and geometric/semantic similarity, and its dataset is the 366patterns in Flat-Folder’s `examples/instagram/`. A full reference has to be recovered before submission. **COrigami** (Zahavy et al., 2026) has a *citation missing* as well: it appears in the metrics survey as the source of flat-foldability as a boolean and a VLM aesthetic score. Same action needed.

Complexity and theory of flat folding. Bern & Hayes (1996) is the citation under the paper’s central claim: geometry does not determine layer order, and deciding overlap order is NP hard even given a valid mountain-valley assignment. Arkin et al. (2004) defines simple foldability; the action space comes from here. Akitaya et al. (2019) covers the all-layers model, which corresponds to sheet-metal bending. Akitaya et al. (2024) is the Flat-Folder paper; deciding a global flat-folded state is NP hard. See also Simple Folding of Mixed Orthogonal CPs, Layer Ordering Constraints and Graphs of Origami Flat Foldings. Flat Origami is

Turing Complete is a defensive citation only, per Section 8. Demaine et al. (2004) separates folded state from folding motion: reachability is free, so the hard question is the discrete step structure; its predecessor is Demaine & Mitchell (2001).

The crease-pattern-to-sequence problem itself. Akitaya et al. (2013) named the problem, by the group whose tools the field now uses: reflection paths, graph rewriting, step graphs; a frog base explodes to 22,665 nodes and 30 minutes, and the authors’ own future work asks for the priority heuristic this paper’s models are being asked to supply. Creasy is an open-source implementation of the above and is usable as a symbolic baseline. [TO RUN if used]

Counting folded states. OEIS Foundation Inc. (2026) lists the foldings of a strip of stamps: 1, 2, 6, 16, 50, 144, 462, 1392, 4536, 14060, ... There is no closed form. See also A001011 and A001416, and Lucas (1891); Koehler (1968); Lunnon (1971); Foldings and Meanders; meanders and stamp foldings are the same combinatorial object.

Spatial reasoning context. Spatial Reasoning in MLLMs surveys the area. Visualization-of-Thought names the mental-imagery question this benchmark can turn into a measurement.

Tools, formats and data. Flat-Folder decides and enumerates flat-folded states and is the source of the four constraint classes cited in Section 6.3. FOLD file format is the de facto standard used by every artifact we ship. Mitani is a crease-pattern editor with folded-form estimation. PurelandFold is not used as training or evaluation data here; it is cited as the closest existing sequence-level corpus and as FoldingAgent’s data.

Statistical physics of random flat-foldability. Optional support for the claim that multiple valid states is the normal case rather than a curiosity, to be included only if Section 4.1 needs reinforcement: Spin Model for Flat-Foldability and Random Locally Flat-Foldable Origami.

Deliberately not cited. Thick folding (Ku & Demaine 2016), bar-and-hinge mechanics, MERLIN2, SWOMPS, Sim-FAST-PY, and the creased-sheet mechanics literature all belong to the thickness and mechanics line of work. They are a separate paper with a verifier several orders of magnitude more expensive, and citing them here would invite a reviewer to ask why this paper does not do that one.